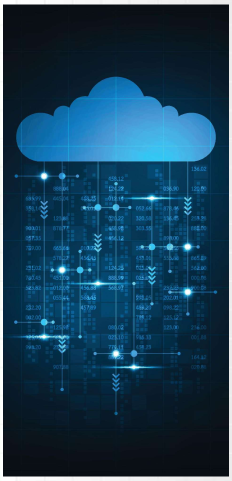

Cloud programming

Rafał Palak

Wrocław University of Science and Technology

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Agenda

- Grading Rules
- Course Outline
- Basic Definitions
- Largest Cloud Providers

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Office hours

- Wed 15-17
- Thu 9-11
- wojciech.thomas@pwr.edu.pl

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Grading rules [1]

- Test

Points	Grade
[0%,50%]	2
(50%,60%]	3
(60%,70%]	3.5
(70%,80%]	4
(80%,90%]	4.5
(90%,100]	5

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Grading rules[2]

- Exemption from the test:
  ?

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Lecture plan [1]

1. Introduction to the subject matter
2. Basic cloud services
3. Docker and Packer
4. Cloud security principles
5. IaC (Infrastructure as Code) tools
6. Data storage in the cloud (files and databases)
7. Serverless architecture and its application
8. Project and implementation of a cloud application

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Lecture plan [2]

9. Tools for continuous integration
10. Best practices in cloud solutions
11. Test

Basic definitions

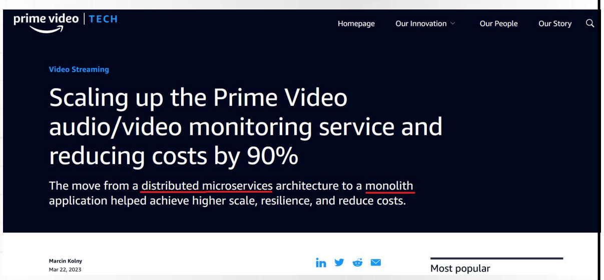
Cloud computing - definition

- Is the **on-demand delivery of IT resources** over the Internet with **pay-as-you-go pricing**. Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases, on an **as-needed basis** from a cloud provider[1]
- Is the **delivery of computing services**—including servers, storage, databases, networking, software, analytics, and intelligence—over the internet ("the cloud") to **offer faster innovation, flexible resources, and economies of scale**. You typically **pay only for cloud services you use**, helping you lower your operating costs, run your infrastructure more efficiently, and scale as your business needs change.[2]
- Is the **on-demand availability of computing resources** (such as storage and infrastructure), as services over the internet. It **eliminates the need for individuals and businesses to self-manage physical resources themselves**, and only **pay for what they use.**[3]

1. https://aws.amazon.com/what-is-cloud-computing/
2. https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-cloud-computing
3. https://cloud.google.com/learn/what-is-cloud-computing

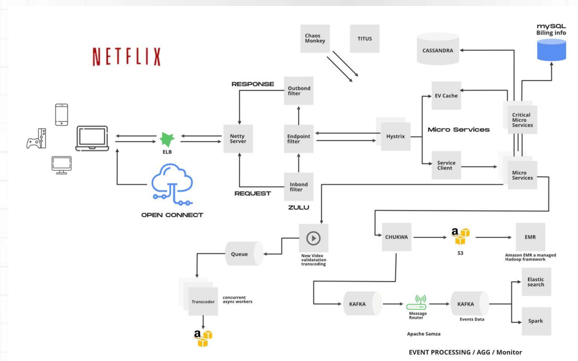
Cloud computing – why?

Cloud computing – scalability

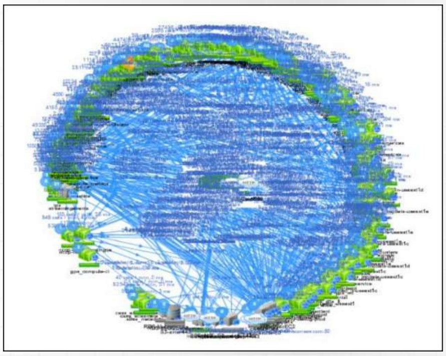


https://www.primevideotech.com/video-streaming/scaling-up-the-prime-video-audio-video-monitoring-service-and-reducing-costs-by-90

Cloud computing – scalability



Cloud computing – scalability



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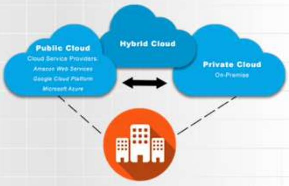
Public Cloud

- Parts of the cloud that are owned and managed by companies that provide access to computing resources to the general public.
- Users do not have to purchase software, hardware, or even infrastructure.
- Managed and maintained by the provider, who may charge a nominal fee or none at all.



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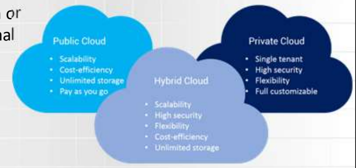
Private Cloud

- A part of the cloud that is intended only for a specific organization, exclusively for its use.
- Maintenance may be performed internally by the organization or even outsourced to an external service provider.



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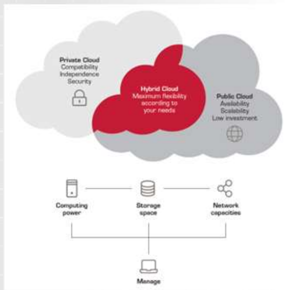
Hybrid Cloud

- It means using private cloud infrastructure along with cloud services that are public in nature.
- The private cloud cannot exist by itself.
- Most companies that have a private cloud configuration access public cloud resources to perform various daily tasks.



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AWS

- Amazon Web Services
- One of the first cloud providers

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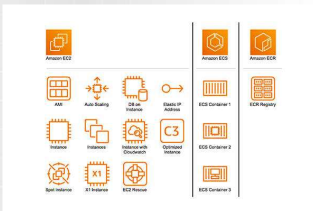
Amazon EC2 (Elastic Cloud Compute)

- Allows users to rent virtual computers on which they can run their own applications.
- EC2 encourages scalable application deployment.
- Users can create, launch, and terminate server instances as needed.
- Pay per second.
- Provides users with control over the geographic location of instances.



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S3 Amazon Simple Storage Service

- Provides storage through a web service interface.
- Can store any type of object up to 5 TB each.
- Each object is stored in a bucket.
- Standardized REST and SOAP interfaces.
- The default protocol is HTTP.

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Cloud providers

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Top 10 Cloud providers [1]

#	Cloud Service Provider	Regions	Availability Zones
1	Amazon Web Services (AWS)	26	84
2	Microsoft Azure	60	116
3	Google Cloud Platform (GCP)	34	103
4	Alibaba Cloud	27	84
5	Oracle Cloud	38	46
6	IBM Cloud (Kyndryl)	11	29
7	Tencent Cloud	21	65
8	OVHcloud	13	33
9	DigitalOcean	8	14
10	Linode (Akamai)	11	11

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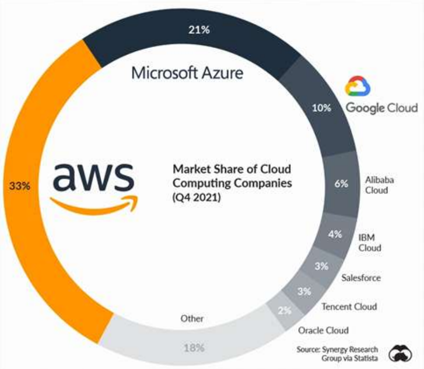
Top 10 Cloud providers [2]



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Prices of services - 2vCPU, 4GB RAM, 80GB SSD

#	Cloud Service Provider	Cost per Month	% Higher
1	Alibaba Cloud	$48.42	–
2	Google Cloud Platform (GCP)	$63.38	31%
3	Microsoft Azure	$70.05	45%
4	Amazon Web Services (AWS)	$71.47	48%

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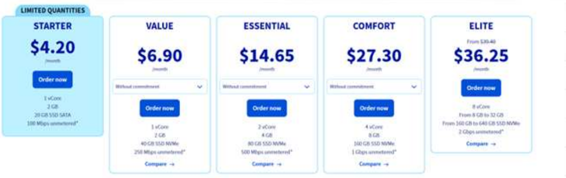
Prices – OVH VPS



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Cloud vs VPS

Feature	VPS	Cloud
Hosted on a single physical server	✓	×
Hosted across multiple physical servers	×	✓
Exceptionally high level of security	✓	✓
Excellent customization options	✓	×
Basic customization options	×	✓
Dedicated resources	✓	
Unlimited resources	×	✓
High level of scalability	×	✓
Guaranteed ability to cope with traffic surges	×	✓
High availability	×	✓

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What to choose?

Cloud

- You have extensive technological knowledge and can easily customize cloud servers.
- Your application experiences both predictable and unpredictable traffic spikes.
- Your application is growing, and you need to ensure that your hosting can handle the additional demand without downtime or interruptions.

VPS

- Your application is relatively small, and the monthly network traffic is predictable (no large spikes in traffic).
- You possess sufficient technical knowledge necessary to customize your VPS.
- You are concerned about malicious attacks on your site.

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AWS

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Azure

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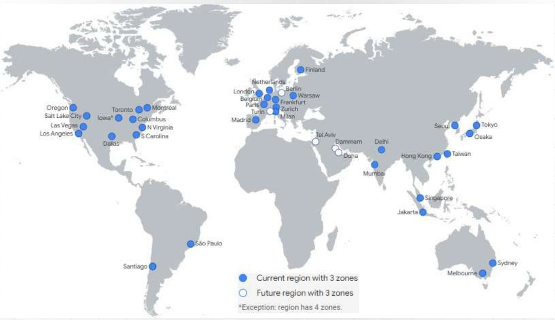
Google Cloud



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Alibaba Cloud

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Oracle Cloud

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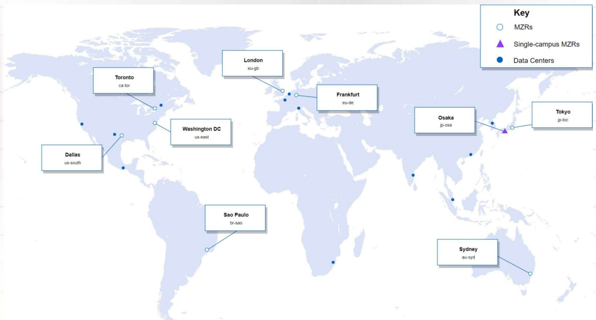
IBM Cloud (Kyndryl)



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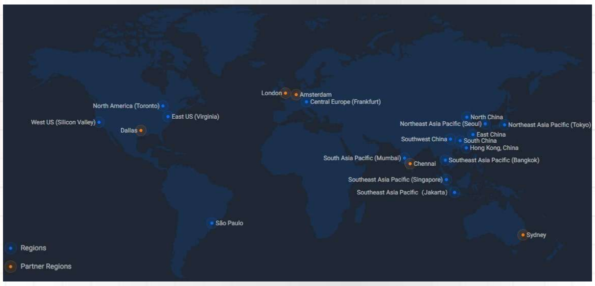
Tencent Cloud



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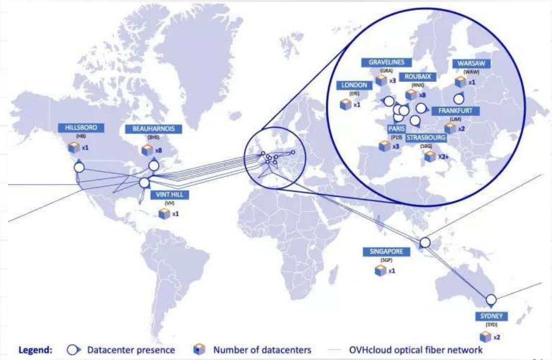
OVHcloud



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DigitalOcean

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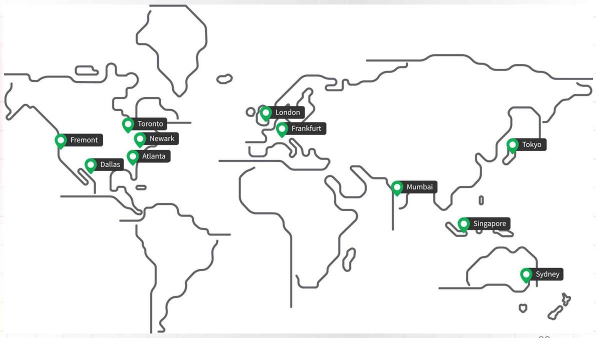
Linode (Akamai)



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Thank you for your attention

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